The A_6 Morin Chern-Positivity Problem

Exact reduction, certified computations, new infinite sectors, kernel lifts, and the remaining proof obligation

A self-contained research and proof-status dossier

31 July 2026

Abstract

For the Morin singularity A_d, Rimányi's conjecture predicts nonnegative coefficients of the Thom polynomial in the ordinary monomial basis of the relative Chern classes. This report studies $d = 6$ using the Bérczi–Szenes iterated-residue formula. The geometric numerator Q_6 is reconstructed exactly; its component multiplicities, chamber normalization, and ballot-orbit interpretation are certified. The colleague-supplied all-level proof for $d \leq 5$ is audited and used only at the precise boundary where it applies.

The full A_6 theorem is *not* proved here, and no complete-orbit counterexample is found. The exact missing proposition is the coefficientwise nonnegativity of one explicit rational series $\mathcal{E}_6$. We prove that this proposition together with the established A_5 theorem and a new singleton-tail theorem would imply every A_6 Chern coefficient is nonnegative. We also prove several infinite faces, three passive recurrences, a positive dominant residue, an eventual one-sided tail, and a four-parameter family of universal A_6 residue-kernel lifts. Exact counterexamples show why raw Laurent positivity, positivity of the kernel-modified representative, pointwise two-state cones, and intrinsic unmarked geometric interpretations of the marked pair packets cannot complete the proof.

Every assertion is labelled as a published input, an exact mathematical proof, an exact computer-assisted certificate, finite evidence, or an open problem. In particular, a large nonegative computation is never used as theorem evidence.

Contents

1 Executive conclusion and epistemic status

1.1 The answer in one paragraph

The full A_6 case remains open in this dossier. The work below does not contain a proof of all A_6 Chern-monomial coefficients and does not contain a negative complete ballot-orbit sum. It does contain a complete exact reduction to one canonical-gauge coefficientwise inequality:

$$\boxed{\mathcal{E}_6(x,y,c,d,e) \in \mathbb{Z}_{\geq 0}[[x,y,c,d,e]].}$$

The series $\mathcal{E}_6$ is defined in Section 9. If this boxed statement is proved, then the full A_6 theorem follows by Theorem 9.2. At present the c^1, c^2d, c^2d^2, and complete de-ray faces are proved, as are several exact active and passive tails, but the joint signed-head domination needed for the boxed statement is open.

1.2 Status vocabulary

Label	Meaning
Published	A theorem taken from a cited archival source.
Imported	A theorem in the colleague-supplied A_5 report. Its load-bearing algebra and contour identities were reconstructed here, but it is not represented as a new peer-reviewed theorem.
Exact	A symbolic identity or mathematical proof valid in all degrees.
Certified	A finite algebraic claim checked with exact integer/rational polynomial arithmetic and an independently described certificate.
Finite	An exhaustive result in a stated finite box. It is a falsifier search or regression test, not an infinite proof.
Open	A proposition neither proved nor disproved in this project or in the primary literature located by the search described below.

1.3 The five-step programme

Step 1. Certify Q_6. Complete. The 22-coordinate associativity scheme, its two top components, their multiplicity one, and the resulting 418-term degree-seven multidegree are exact.

Step 2. Build independent coefficient and orbit engines. Complete. A factorwise chamber engine and a global denominator-recurrence engine agree exactly in cross-check boxes. The ballot-orbit and 0-Hecke operators are independently verified.

Step 3. Transfer the proved A_5 information without assuming a false stronger statement. Complete. The $m = 0$ block of each complete A_6 orbit is a complete A_5 orbit after deletion of a terminal zero. Full A_5 packet positivity is sufficient there; A_5 first-pair positivity is not assumed.

Step 4. Prove infinite A_6 sectors and reduce the open interior. Partly complete. The first-swap singleton sector, three all-degree passive faces, three tail recurrences, a positive dominant residue, slice-wise eventual positivity, and a universal kernel lift are proved. The canonical pair-extension series $\mathcal{E}_6$ remains open.

Step 5. Close the theorem or isolate the irreducible obstruction. Not complete. The exact obstruction is now the joint domination of a positive $x = 1/3$ residue over a signed active head along the *cd* and related recession rays. Several natural mechanisms are proved incapable of supplying that domination.

2 The problem, from first principles

2.1 Map germs, Morin singularities, and relative Chern classes

Definition 2.1 (Map germ and relative dimension)**.** Let $f : (\mathbb{C}^n, 0) \to (\mathbb{C}^{n+\ell}, 0)$ be a holomorphic map germ. The nonnegative integer ℓ is its *relative dimension*. For a global map $f : X^n \to Y^{n+\ell}$, the virtual normal bundle is $f^*TY - TX$.

Definition 2.2 (Relative Chern classes)**.** The total relative Chern class is

$$c(f) = c(f^*TY - TX) = \frac{c(f^*TY)}{c(TX)} = 1 + c_1 + c_2 + \cdots .$$

We use $c_0 = 1$ and $c_j = 0$ for $j < 0$. These conventions are load-bearing in every residue and orbit formula.

Definition 2.3 (Morin singularity)**.** The Morin contact singularity A_d is the corank-one singularity whose unital local algebra is

$$A_d = \mathbb{C}[t]/(t^{d+1}).$$

The complex codimension of its locus in relative dimension ℓ is $d(\ell + 1)$.

Definition 2.4 (Thom polynomial)**.** For a sufficiently generic holomorphic map, the cohomology class dual to the closure of its A_d-locus is obtained by substituting the relative Chern classes into a universal polynomial $\mathrm{Tp}^{\ell}_{A_d}$. With $L = \ell + 1$, it is homogeneous of weighted degree dL, where c_j has weight j.

Definition 2.5 (Chern-monomial basis)**.** A partition $\lambda = (\lambda_1 \geq \cdots \geq \lambda_s > 0)$ of dL defines $c_\lambda = \prod_{i=1}^{s} c_{\lambda_i}$. We may pad λ by zeros to length d, because $c_0 = 1$.

Conjecture 2.6 (Rimányi's Chern-monomial positivity)**.** *For every $d \geq 1$, $L \geq 1$, and partition $\lambda \vdash dL$,*

$$[c_\lambda] \, \mathrm{Tp}^{L-1}_{A_d} \geq 0.$$

The present report concerns $d = 6$.

2.2 Three positivity notions that must not be conflated

1. *Schur positivity* is a geometric theorem for stable complex singularity classes in the relevant convention. A Schur-positive polynomial may have negative coefficients after conversion to the ordinary c_λ basis.

2. *Chern-monomial positivity* is Conjecture 2.6. Formally it is positivity in the elementary-symmetric generators c_j.

3. *Fixed-chamber Laurent positivity* asks every coefficient of one ordered Bérczi–Szenes Laurent expansion to be nonnegative. It implies Chern positivity, but is presentation-dependent and is false already for A_5.

Thus the valid implication is

$$\text{fixed-chamber Laurent positivity} \Longrightarrow \text{Chern-monomial positivity} \Longrightarrow \text{Schur positivity},$$

and neither reverse implication is available.

2.3 A fully worked toy example: A_2

Example 2.7. For $d = 2$, put $a = z_1/z_2$. The chamber series is

$$F_2(a) = \frac{1-a}{1-2a} = 1 + \sum_{r\geq 1} 2^{r-1}a^r.$$

A monomial a^r corresponds to the zero-sum charge $\alpha = (r, -r)$. At level $L = \ell + 1$, its Chern monomial is $c_{L+r}c_{L-r}$; it vanishes if $r > L$. Therefore

$$\mathrm{Tp}_{A_2}^{L-1} = c_L^2 + \sum_{r=1}^{L} 2^{r-1} c_{L-r}c_{L+r}.$$

At $L = 2$, for instance,

$$\mathrm{Tp}_{A_2}^{1} = c_2^2 + c_1c_3 + 2c_4,$$

because $c_0c_4 = c_4$. This example displays the complete mechanism used for A_6: chamber exponent $\to$ zero-sum charge $\to$ shifted Chern indices. What changes at A_6 is that several ballot orderings of the same charge multiset contribute to one commutative Chern monomial and must be summed.

3 Literature map and the exact novelty boundary

3.1 Foundational results

Rimányi's 2001 paper introduced restriction-equation and incidence methods and stated the ordinary Chern-monomial positivity conjecture for Morin singularities. Pragacz and Weber proved Schur-positivity results for Thom polynomials of stable complex singularity classes. Fehér and Rimányi later developed stable Thom series for contact singularities, which explain why the all-ℓ family is governed by one shifted formal series.

Bérczi and Szenes proved an all-relative-dimension iterated-residue formula for Morin Thom polynomials. Their formula makes A_d explicitly computable once the Borel-orbit multidegree Q_d is known. It also led to a stronger fixed-chamber coefficient-positivity conjecture. The latter is false from $d = 5$ onward; this does not settle Rimányi's weaker, intrinsic conjecture.

Bérczi's later non-reductive-GIT work gives a toric blow-up route to the same residue data and relates curvilinear Hilbert schemes to invariant jet differentials. The published algorithm does not provide the exact A_6 terminal leaf table, so it is not presently an independent executable Q_6 certificate.

3.2 What the search did and did not find

The literature search covered the primary Bérczi–Szenes formula, Rimányi's foundational paper and current primer, the Thom-series and Schur-positivity papers, the curvilinear Hilbert-scheme model, and the non-reductive-GIT formula. No archival source located by that search proves or disproves ordinary Chern-monomial positivity for all A_6 levels. The complete A_5 proof used below comes from the supplied 15-page colleague report dated 29 July 2026 and is explicitly labelled "imported," not "published."

3.3 Safe novelty claims

Subject to a broader priority search, the following appear new relative to the primary sources inspected:

- the public exact 418-term Q_6 reconstruction with a scheme-theoretic multiplicity audit;
- the all-degree first-swap singleton theorem;
- the exact reduction of full A_6 positivity to the canonical $\mathcal{E}_6$ series plus the full A_5 boundary block;
- the reduced 25-factor pair transfer, passive recurrences, and positive $x = 1/3$ residue;
- the four-parameter universal A_5-kernel lift into A_6; and
- the gauge-dependence obstruction showing that marked first-S_2 packets are not intrinsic unmarked characteristic numbers.

These are mathematical novelty assessments, not unconditional assertions of priority.

4 The Bérczi–Szenes residue and packet formula

4.1 The universal rational function

Let

$$\Delta_d(z) = \prod_{1\le i<j\le d} (z_i - z_j)$$

and

$$D_d(z) = \prod_{\ell=2}^{d} \prod_{m=1}^{\ell-1} \prod_{r=1}^{\min(m,\ell-m)} (z_m + z_r - z_\ell).$$

The Bérczi–Szenes theorem gives

$$\mathrm{Tp}^{\ell}_{A_d} = (-1)^d \,\mathrm{IRes}_{\mathcal{C}} \frac{\Delta_d(z)Q_d(z)}{D_d(z)} \prod_{i=1}^{d} c(1/z_i) z_i^{\ell}\, dz_i, \tag{4.1}$$

in the chamber $\mathcal{C} : |z_1| \ll |z_2| \ll \cdots \ll |z_d|$. Our convention is $\mathrm{Res}_{z=\infty} f(z)\, dz = -[z^{-1}]f(z)$. The d residue signs therefore cancel the outer factor $(-1)^d$.

Define the fixed-chamber Laurent series

$$K_d(z) = \mathcal{E}_{\mathcal{C}}\left(\frac{\Delta_d(z)Q_d(z)}{D_d(z)}\right) = \sum_{\substack{a\in\mathbb{Z}^d\\ \sum_i a_i=0}} k_d(a)z^a. \tag{4.2}$$

Theorem 4.1 (Distinct-orbit packet formula)**.** *Let $L = \ell + 1$, and pad a partition $\lambda \vdash dL$ by zeros to a d-tuple. Then*

$$[c_\lambda]\,\mathrm{Tp}_{A_d}^{L-1} = \sum_{\mu\in\mathrm{Orb}_{\mathrm{dist}}(\lambda)} k_d(\mu - L\mathbf{1}). \tag{4.3}$$

If $\mathrm{len}(\lambda) > d$*, the coefficient is zero.*

Structured proof.

$\langle 1\rangle 1.$ Expand one term of the integrand. A Laurent monomial $k_d(a)z^a$, multiplied by $\prod_i c_{j_i} z_i^{-j_i} z_i^{L-1}$, has z_i-exponent $a_i - j_i + L - 1$.

$\langle 1\rangle 2.$ Determine when the residue selects it. The z_i^{-1} condition is $a_i - j_i + L - 1 = -1$, hence $j_i = L + a_i$.

$\langle 1\rangle 3.$ Check the total Chern weight. Because $\sum_i a_i = 0$, $\sum_i j_i = dL$.

$\langle 1\rangle 4.$ Fix a commutative Chern monomial. To obtain c_λ, the labelled tuple $(j_1, \ldots, j_d)$ must be a distinct permutation of the padded λ.

$\langle 1\rangle 5.$ Audit stabilizers. Equal parts of λ do not label distinct assignments. Therefore one sums over distinct permutations and introduces no stabilizer factorial.

$\langle 1\rangle 6.$ Audit signs and invalid indices. The outer and residue-at-infinity signs cancel. Assignments with $j_i < 0$ vanish by $c_{j_i} = 0$. This gives (4.3). □

4.2 Universal residue kernel

For a coefficient array f on the zero-sum lattice, define

$$\epsilon_+(f)([a]) = \sum_{b\in\mathrm{Orb}_{\mathrm{dist}}(a)} f(b).$$

The packet theorem implies that a homogeneous numerator correction P changes no Thom polynomial at any level exactly when

$$\epsilon_+\mathcal{E}_{\mathcal{C}}\left(\frac{\Delta_d P}{D_d}\right) = 0. \tag{4.4}$$

At the abstract array level,

$$\ker\epsilon_+ = \sum_{i=1}^{d-1}(1 - s_i)\mathbb{Z}^{\Lambda_0}, \tag{4.5}$$

where s_i exchanges adjacent coordinates. Equation (4.5) describes the kernel after expansion. Constructing a rational numerator whose fixed-chamber expansion realizes a desired kernel array is substantially harder because contour safety is not automatic.

5 Exact reconstruction of Q_6

5.1 The 22-coordinate associativity scheme

Use coordinates $u^\ell_{mr} = u^\ell_{rm}$ with $1 \le m \le r$ and $m + r \le \ell \le 6$. There are 22 such coordinates, and u^ℓ_{mr} has torus weight $z_m + z_r - z_\ell$. For $i + j + m \le \ell$, associativity equates

$$\sum_{s=j+m}^{\ell-i} u^s_{jm} u^\ell_{is}, \qquad \sum_{s=i+m}^{\ell-j} u^s_{im} u^\ell_{js}, \qquad \sum_{s=i+j}^{\ell-m} u^s_{ij} u^\ell_{ms}. \tag{5.1}$$

Removing equal and zero expressions leaves ten relations. Let I_{basic} be their ideal.

Theorem 5.1 (Certified component decomposition)**.** *Over $\mathbb{Q}$, the basic ideal has*

$$I_{\text{basic}} = P_1 \cap P_2,$$

where P_1, P_2 are distinct prime ideals of dimension 15*. The ideal is radical and both components occur with multiplicity one. The unwanted coordinate component is*

$$P_1 = \langle u^2_{11}, u^3_{12}, u^3_{11}, u^5_{14}, u^6_{14}, u^6_{15}, u^6_{24} \rangle. \tag{5.2}$$

Structured proof.

$\langle 1 \rangle$1. Construct all 22 variables and the ten relations from (5.1), rather than transcribing an already simplified ideal.

$\langle 1 \rangle$2. Add the quartic relation printed for the non-linear component in the Bérczi–Szenes $d = 6$ calculation. Every term has torus weight $2z_1 + 3z_2 + 3z_3 - 2z_4 - z_5 - z_6$.

$\langle 1 \rangle$3. Exact Singular computations over $\mathbb{Q}$ certify $\dim P_1 = \dim P_2 = 15$, primality of each, and $I_{\text{basic}} = P_1 \cap P_2$.

$\langle 1 \rangle$4. An intersection of distinct primes is radical. Its fundamental cycle is the sum of the two prime components with multiplicity one. ☐

Remark 5.2. The ideal certificate proves the decomposition and multiplicities. Identifying $V(P_2)$ with the desired Borel-orbit closure still uses the containment, irreducibility, and dimension argument in the published Bérczi–Szenes geometry.

5.2 The multidegree certificate

A Gröbner degeneration of I_{basic} in degree-reverse-lexicographic order has 52 height-seven coordinate covers. Forty-eight have localized length one and four have length two, for total multiplicity 56. Summing the products of the seven corresponding torus weights gives the multidegree of the basic scheme. Subtracting the P_1 coordinate component exactly once gives Q_6.

Theorem 5.3 (Certified Q_6 data)**.** *The resulting polynomial Q_6 has total degree* 7*, 418 nonzero terms, coefficient range* $[-233, 161]$*, and canonical SHA-256*

e92f914774931f3e45559103b2bec4bc4317c5e6b88956f44766 48ea156fea83.

The degree, reverse-lexicographic, and lexicographic term orders give the same polynomial. It also satisfies $[z_6^4]Q_6 = Q_5$ and agrees, after the documented sign convention, with the Thom Polynomial Portal.

Warning 5.4. The hash is a transcription and reproducibility checksum, not a proof by itself. The proof-bearing objects are the exact ideal decomposition, flatness of the Gröbner degeneration, localized multiplicity calculations, and subtraction of P_1 with certified multiplicity one.

6 Chambervariables, ballot charges, and full S_6 folding

Normalize $z_6 = 1$, set $x_j = z_j/z_{j+1}$, and put $I_{r,s} = x_r \cdots x_{s-1} = z_r/z_s$. The normalized chamber series is

$$F_6(x_1, \dots, x_5) = \widehat{Q}_6(x) \, \frac{\prod_{1 \le r < s \le 6}(1 - I_{r,s})}{\displaystyle\prod_{\ell=2}^{6} \prod_{m=1}^{\ell-1} \prod_{r=1}^{\min(m,\ell-m)} (1 - I_{m,\ell} - I_{r,\ell})}, \tag{6.1}$$

where the first line means multiplication of $\widehat{Q}_6$ by the displayed fraction. Here

$$\widehat{Q}_6(x) = -\frac{Q_6(z(x))}{x_4 x_5^3}$$

and its constant term is 1.

Write

$$F_6 = \sum_{\beta \in \mathbb{Z}_{\ge 0}^5} A_\beta x^\beta.$$

Set $\beta_0 = \beta_6 = 0$ and

$$\alpha_i = \beta_i - \beta_{i-1} \quad (1 \le i \le 6). \tag{6.2}$$

Then $\sum_i \alpha_i = 0$, and the five proper prefix sums of α are exactly $\beta_1, \dots, \beta_5$. Thus $\beta \ge 0$ is equivalent to α being a ballot ordering of a zero-sum multiset.

Theorem 6.1 (Ballot form of every A_6 coefficient)**.** *For a zero-sum six-multiset M, define*

$$C_6(M) = \sum_{\alpha \in \mathrm{Bal}_6(M)} A_{\beta(\alpha)}. \tag{6.3}$$

If $L + \min M \ge 0$, this is the coefficient of $\prod_{\alpha \in M} c_{L+\alpha}$ in $\mathrm{Tp}_{A_6}^{L-1}$. Conversely every all-level Chern coefficient arises this way. Therefore full A_6 positivity is equivalent to $C_6(M) \ge 0$ for every zero-sum six-multiset M.

Proof. This is Theorem 4.1 after identifying the chamber charge a with $\alpha(\beta)$. The ballot condition records precisely which permutations occur in the prescribed chamber. □

6.1 The 0-Hecke fold

For $1 \le i \le 5$, let s_i swap α_i, α_{i+1}. In prefix coordinates it changes only

$$(s_i\beta)_i = \beta_{i-1} + \beta_{i+1} - \beta_i. \tag{6.4}$$

The dominance gap is

$$D_i(\beta) = \alpha_i - \alpha_{i+1} = 2\beta_i - \beta_{i-1} - \beta_{i+1}. \tag{6.5}$$

Define Π_i to merge the two coefficients in a strict s_i-orbit into the member with $D_i > 0$, and to retain a wall coefficient once. These compare-and-merge operators satisfy

$$\Pi_i^2 = \Pi_i, \qquad \Pi_i\Pi_{i+1}\Pi_i = \Pi_{i+1}\Pi_i\Pi_{i+1}, \qquad \Pi_i\Pi_j = \Pi_j\Pi_i \ (|i-j| > 1). \tag{6.6}$$

Applying

$$(1, 2, 1, 3, 2, 1, 4, 3, 2, 1, 5, 4, 3, 2, 1) \tag{6.7}$$

produces exactly $C_6(M)$ at the decreasing ordering of M. Wall retention is what prevents over-counting repeated charges.

6.2 Exact finite evidence and the known uniform sector

Exact full-orbit computation finds no negative coefficient for $\ell = 0, \ldots, 8$. These are complete finite theorems at the stated levels only. A maximum-charge enumeration gives positive complete orbit sums through $p = \max M = 6$: the $p = 6$ layer has 1,226 orbits, minimum 2,400, maximum 19,377,586,240, and total mass 495,357,900,928.

The colleague report also supplies an all-rank geometric theorem: if the largest Chern index obeys $\lambda_1 \leq L + 1$, equivalently $\max M \leq 1$, then the coefficient is nonnegative. Its proof descends a curvilinear characteristic number to the plane and interprets the result through Jucys–Murphy transposition words. This establishes the uniform $p \leq 1$ sector. The open A_6 complement begins at $p = 2$.

7 The imported A_5 theorem and its exact scope

The supplied colleague report proves all-level A_5 Chern-monomial positivity. Its decisive factorization uses

$$B = 2z_1 - z_2, \quad F = 2z_1 - z_3, \quad E = z_1 + z_2 - z_3,$$
$$D = 2z_2 - z_4, \quad M = z_2 + z_3 - z_5, \quad C = z_1 + z_4 - z_5,$$
$$G = 2z_1 + z_2 - z_5, \quad J = 2z_1 + z_3 - z_5$$

and

$$Q_5 = G(DJ + BC) = GDJ + GBC. \tag{7.1}$$

It proves GBC is universally residue-null using two chamber-safe contour swaps, and gives a coefficientwise-positive Stanley decomposition for the fixed-chamber series of GDJ. Thus full A_5 Chern positivity follows even though the original Q_5 chamber series has negative coefficients.

Warning 7.1 (What A_5 does not automatically supply). Full A_5 packet positivity does not imply that every first-S_2 subpacket of the original Q_5 series is nonnegative. A first pair is only part of a complete orbit and can be repaired by later permutations. The A_6 argument below therefore uses full A_5 positivity only after regrouping the complete $m = 0$ boundary.

8 First-swap singletons: a complete infinite theorem

Write

$$F_6(a,b,c,d,e) = \sum_{i,j,k,\ell,m\geq 0} A_{i,j,k,\ell,m} a^i b^j c^k d^\ell e^m. \tag{8.1}$$

Swapping the first two charges sends the prefix tuple to

$$\tau(i,j,k,\ell,m) = (j-i,j,k,\ell,m). \tag{8.2}$$

If $i>j$, then $j-i<0$, so the reflected ordering is not ballot. Such a state is a first-swap singleton.

Theorem 8.1 (First-swap singleton theorem)**.** *For every $i,j,k,\ell,m \geq 0$,*

$$i>j \quad\Longrightarrow\quad A_{i,j,k,\ell,m} \geq 0. \tag{8.3}$$

Structured proof.

⟨1⟩1. Isolate the only denominator creating an arbitrarily long first-coordinate tail. Put

$$H(a) = \frac{1-a}{1-2a}, \qquad J_6(a,b,c,d,e) = \frac{1-2a}{1-a} F_6(a,b,c,d,e). \tag{8.4}$$

Then $F_6 = HJ_6$.

⟨1⟩2. Prove the triangular support of J_6. After removing $1-a$ from the Vandermonde, every numerator monomial of J_6 satisfies $\deg_a \leq \deg_b$. Every remaining denominator transition has the same property because any interval monomial containing a also contains b. Products preserve the inequality. Hence, for fixed j,k,ℓ,m, the a-degree of $[b^j c^k d^\ell e^m]J_6$ is at most j.

⟨1⟩3. Use the geometric tail of H. Since

$$H(a) = 1 + \sum_{r\geq 1} 2^{r-1} a^r,$$

and $i>j$, every convolution term selecting a^i lies in the geometric part. Thus

$$A_{i,j,k,\ell,m} = 2^{i-1}[b^j c^k d^\ell e^m] J_6(1/2,b,c,d,e). \tag{8.5}$$

⟨1⟩4. Factor the specialization exactly. Set $B=b/2$ and $t=cde$. Exact cancellation gives

$$\begin{aligned} J_6(1/2,b,c,d,e) = G(B,t) &\frac{1-c}{1-c-Bc}\frac{1-Bc}{1-3Bc}\frac{1-d}{1-d-Bcd} \\ &\times \frac{1-cd}{1-cd-Bcd}\frac{1-Bcd}{1-3Bcd}\frac{1-e}{1-e-Bt}\frac{1-de}{1-de-Bt}, \end{aligned} \tag{8.6}$$

where all eight displayed factors are multiplied and

$$G(B,t) = \frac{(1-B)(1-t)(1-Bt)(1-t-3Bt)}{(1-3B)(1-2t)(1-t-Bt)(1-3Bt)}. \tag{8.7}$$

⟨1⟩5. Prove the seven simple ratios nonnegative. Each is of the form

$$\frac{1-u}{1-u-v} = 1 + \frac{v}{1-u-v}, \tag{8.8}$$

or $(1-u)/(1-3u) = 1 + 2u/(1-3u)$. Here u, v are monomials with nonnegative coefficients and zero constant term, so each reciprocal expansion is coefficientwise nonnegative.

$\langle 1\rangle 6$. Decompose the residual core.

$$\begin{aligned} G(B,t) = 1 + \frac{2B}{1-3B} + \frac{t}{1-2t} + \frac{2Bt^2}{(1-2t)(1-3Bt)} \\ + \frac{2Bt(1-t)}{(1-3B)(1-3Bt)(1-t-Bt)}. \end{aligned} \tag{8.9}$$

The first four summands are manifestly nonnegative. In the last one, $(1-t)/(1-t-Bt) = 1 + Bt/(1-t-Bt)$, so it is also nonnegative.

$\langle 1\rangle 7$. Conclude. Equation (8.6) is coefficientwise nonnegative, and the scalar in (8.5) is positive. Hence (8.3) holds in all degrees. □

9 The canonical pair-extension series

9.1 Positive-quadrant first pairs

Put $x = ab$ and $y = b$. If $p, q \geq 0$, the two ballot orientations of the first two nonnegative charges contribute

$$A^{(6)}_{p,p+q,k,\ell,m} + A^{(6)}_{q,p+q,k,\ell,m}. \tag{9.1}$$

For $p = q$, the symmetric generating series counts the fixed word twice; its sign is therefore still the sign of the actual one-word packet.

Let $\mathcal{P}_6(x, y, c, d, e)$ be the series whose coefficient is (9.1), and let $\mathcal{P}_5(x, y, c, d)$ be the analogous original Q_5 series. Exact stabilization gives

$$F_6(a, b, c, d, 0) = F_5(a, b, c, d), \qquad \mathcal{P}_6(x, y, c, d, 0) = \mathcal{P}_5(x, y, c, d). \tag{9.2}$$

Define

$$\boxed{\mathcal{E}_6(x, y, c, d, e) = \frac{\mathcal{P}_6(x, y, c, d, e) - \mathcal{P}_5(x, y, c, d)}{e}.} \tag{9.3}$$

Then

$$[x^p y^q c^k d^\ell e^m]\mathcal{E}_6 = A^{(6)}_{p,p+q,k,\ell,m+1} + A^{(6)}_{q,p+q,k,\ell,m+1}. \tag{9.4}$$

9.2 The exact rational operator

Put

$$T(a, b, c, d, e) = \frac{1}{e}\left(\frac{1-2a}{1-a}F_6(a, b, c, d, e) - \frac{1-2a}{1-a}F_5(a, b, c, d)\right) \tag{9.5}$$

and $\mathcal{T}(x, y, c, d, e) = T(x/y, y, c, d, e)$. The support inequality $\deg_a \leq \deg_b$ makes this a positive-quadrant formal series. Define

$$R(x, y, c, d, e) = \frac{(y-x)\mathcal{T}(x, y, c, d, e) - x\mathcal{T}(x, 2x, c, d, e)}{y - 2x}. \tag{9.6}$$

The numerator vanishes at $y = 2x$, so the apparent pole is removable. Exact polynomial division gives

$$\boxed{\mathcal{E}_6 = R(x, y, c, d, e) + R(y, x, c, d, e).} \tag{9.7}$$

Lemma 9.1 (Coefficient action of the removable division)**.** *If $\mathcal{T} = \sum t_{i,j} x^i y^j$, set $w_0 = w_1 = 1$ and $w_r = 2^{r-1}$ for $r \geq 2$. Then*

$$[x^p y^q]R = \sum_{i=0}^{p} w_{p-i} t_{i,p+q-i}. \tag{9.8}$$

Proof. Polynomial division gives

$$\frac{(y-x)y^j - 2^j x^{j+1}}{y - 2x} = \sum_{r=0}^{j} w_r x^r y^{j-r}.$$

Multiply by $t_{i,j} x^i$, sum, and extract $x^p y^q$. □

9.3 The exact completion theorem

Theorem 9.2 (Conditional completion of A_6)**.** *Assume the imported full A_5 packet theorem, the proved singleton Theorem 8.1, and*

$$\mathcal{E}_6 \in \mathbb{Z}_{\geq 0}[[x, y, c, d, e]]. \tag{9.9}$$

Then every coefficient of $\mathrm{Tp}^{\ell}_{A_6}$ in the ordinary Chern-monomial basis is nonnegative for every $\ell \geq 0$.

Structured proof.

$\langle 1 \rangle 1.$ Fix a zero-sum six-multiset M. By Theorem 6.1, it suffices to prove $C_6(M) \geq 0$.

$\langle 1 \rangle 2.$ Split by the final chamber exponent. For a ballot word α, $\alpha_6 = -\beta_5$. Hence

$$\beta_5 = 0 \iff \alpha_6 = 0, \qquad \beta_5 > 0 \iff \alpha_6 < 0. \tag{9.10}$$

Write $C_6(M) = B_0(M) + B_+(M)$ for these two disjoint blocks.

$\langle 1 \rangle 3.$ Identify the complete boundary block. If $0 \notin M$, then $B_0(M) = 0$. If $0 \in M$, delete the terminal zero from every contributing word. This is a bijection

$$\{\alpha \in \mathrm{Bal}_6(M) : \alpha_6 = 0\} \longleftrightarrow \mathrm{Bal}_5(M \setminus \{0\}). \tag{9.11}$$

It is stabilizer-safe even when M contains repeated zeros because the objects are distinct value-words, not labelled copies.

$\langle 1 \rangle 4.$ Transfer coefficients termwise. By (9.2),

$$A^{(6)}_{i,j,k,\ell,0} = A^{(5)}_{i,j,k,\ell}.$$

Therefore

$$B_0(M) = C_5(M \setminus \{0\}) \geq 0 \tag{9.12}$$

by the full A_5 theorem.

$\langle 1\rangle 5.$ Partition the positive-β_5 block by the first swap. Write $p = \alpha_1 = \beta_1$ and $q = \alpha_2 = \beta_2 - \beta_1$.

$\langle 1\rangle 6.$ Handle two distinct ballot words. If $p, q \geq 0$ and $p \neq q$, both first orders are ballot. Their sum is a coefficient of $\mathcal{E}_6$ by (9.4), and is nonnegative by (9.9).

$\langle 1\rangle 7.$ Handle a fixed word. If $p = q \geq 0$, the corresponding coefficient of $\mathcal{E}_6$ is twice the one-word coefficient. Its nonnegativity is equivalent to the sign of the actual fixed packet.

$\langle 1\rangle 8.$ Handle a singleton. If $q < 0$, the reflected word is not ballot. Since $\beta_2 = p + q \geq 0$, this is exactly the region $\beta_1 > \beta_2$, which is nonnegative by Theorem 8.1.

$\langle 1\rangle 9.$ Sum disjoint nonnegative pieces. The first-swap classes exhaust $B_+(M)$, so $B_+(M) \geq 0$. Together with (9.12), $C_6(M) \geq 0$. Since M was arbitrary, the full A_6 theorem follows. □

Remark 9.3 (The precise invalid stronger inference). Full A_5 packet positivity does not imply $\mathcal{P}_5 \geq 0$. Thus one may not infer $\mathcal{P}_6 = \mathcal{P}_5 + e\mathcal{E}_6 \geq 0$. Theorem 9.2 avoids that error by aggregating the entire $m = 0$ block before using A_5.

10 What is proved about $\mathcal{E}_6$

10.1 A genuinely reduced ordered half

Exact cancellation writes $R = N_R/D_R$, where N_R has 44,866 terms, 22,471 negative terms, multidegree $(15, 9, 21, 17, 9)$, and a factor cd. The denominator has 25 irreducible constant-one factors:

$$\begin{gathered} 1-2x,\ 1-x-y,\ 1-2cx,\ 1-cx-cy,\ 1-2cy,\ 1-c-cx, \\ 1-2cdx,\ 1-cdx-cdy,\ 1-2cdy,\ 1-cd-cdx,\ 1-cd-cdy, \\ 1-d-cdx,\ 1-2cdex,\ 1-cdex-cdey,\ 1-2cdey, \\ 1-cde-cdex,\ 1-cde-cdey,\ 1-2cde, \\ 1-de-cdex,\ 1-de-cdey,\ 1-e-cdex, \\ 1-3x,\ 1-3cx,\ 1-3cdx,\ 1-3cdex. \end{gathered} \tag{10.1}$$

Every reciprocal is coefficientwise nonnegative. Exact rational points on the zero set of each factor show no factor divides N_R. A signed reduced numerator rules out the simplest numerator certificate; it is not a negative series coefficient.

10.2 Proved all-degree faces

Theorem 10.1 (Three complete ordered faces)**.** *The following coefficients of the ordered half are nonnegative for every active bidegree:*

$$\begin{aligned} {}[x^p y^q c^1 d^\ell e^m]R &\geq 0 \quad (p, q, m \geq 0,\ \ell \geq 1), \\ [x^p y^q c^2 d^1 e^m]R &\geq 0 \quad (p, q, m \geq 0), \\ [x^p y^q c^2 d^2 e^m]R &\geq 0 \quad (p, q, m \geq 0). \end{aligned} \tag{10.2}$$

Consequently the same faces of $\mathcal{E}_6 = R + R^\tau$ are nonnegative.

Proof. Exact coefficient extraction decomposes the c^1 face into

$$[c]R = dR_0 + \frac{de}{1-e}R_+ + \frac{d^2e}{1-de}C, \tag{10.3}$$

where

$$\begin{aligned} R_0 &= \frac{(1+y)(1-x)}{1-2x} + \frac{x(2-x)(1-x)}{(1-3x)(1-x-y)}, \\ R_+ &= \frac{x(1-x)}{1-2x} + \frac{x^2(1-x)}{(1-3x)(1-x-y)}, \\ C &= \frac{x^3(1-x)}{(1-2x)(1-3x)(1-x-y)} + \frac{xy(1-x)}{(1-2x)(1-x-y)} + \frac{y(1-x)}{1-2x}. \end{aligned} \tag{10.4}$$

Use $(1-x)/(1-2x) = 1 + x/(1-2x)$ and $(2-x)(1-x)/(1-3x) = 2 + 3x + 10x^2/(1-3x)$. All summands are nonnegative. The c^2d face has an analogous $S_0 + eS_+/(1-e)$ decomposition. For c^2d^2, the coefficients are affine in the e-exponent from e^2 onward, with a nonnegative increment; the two exceptional rows have explicit nonnegative 2^n- and 3^n-formulas. All identities are exact polynomial divisions and are independently regression-tested. □

10.3 Exact passive recurrences

Write $R_{p,q,k,\ell,m} = [x^p y^q c^k d^\ell e^m]R$. The following are exact:

$$\begin{aligned} R_{p,q,k,\ell,m} &= R_{p,q,k,\ell,m-1} + R_{p-1,q,k-1,\ell-1,m-1} && (m > \ell), \\ R_{p,q,k,\ell,m} &= R_{p,q,k-1,\ell,m} + R_{p-1,q,k-1,\ell,m} && (k > p+q+\ell+1), \\ R_{p,q,k,\ell,m} &= R_{p,q,k,\ell-1,m} + R_{p-1,q,k-1,\ell-1,m} && (\ell > k+m+1). \end{aligned} \tag{10.5}$$

An index with a negative component is interpreted as zero. These follow by isolating, respectively, the factors $1-e-cdex$, $1-c-cx$, and $1-d-cdx$, and proving the remaining numerator and transition supports lie on the required side of the corresponding grading hyperplanes.

Warning 10.2. The recurrences do not reduce the problem to finitely many passive indices. Their asymptotic head cone has the three rays $(1,1,0)$, $(0,1,1)$, and $(1,1,1)$ in (k,ℓ,m), corresponding to cd, de, and cde.

10.4 The finite active core is signed

Multiplying R by the ten denominator factors with passive transitions of active degree zero leaves a 15-factor active denominator. For fixed (p,q), the resulting core K vanishes outside

$$k \le p+q+6, \qquad \ell \le p+q+8, \qquad m \le p+q+5. \tag{10.6}$$

Nevertheless

$$K_{0,0,1,1,1} = -1. \tag{10.7}$$

At the origin it repairs through the exact circuit

$$K(0,0) = cd(1-c)(1-cd)^2(1-d)(1-cde)^2(1-de)^2(1-e), \tag{10.8}$$

and division by the passive factors gives

$$R(0,0) = \frac{cd}{1-2cde}. \tag{10.9}$$

This is the model phenomenon: the signed finite core becomes positive only when ratios such as $(1-u)/(1-u-v)$ remain intact.

10.5 Ordered positivity is false

The exact coefficients

$$[x^1y^0c^2d^3e^1]R = -1, \qquad [x^1y^0c^2d^3e^1]R(y,x) = 1 \tag{10.10}$$

cancel in $\mathcal{E}_6$. Thus an ordered proof $R \geq 0$ is impossible even though the symmetric coefficient is zero.

11 The active residue and the exact unbounded obstruction

11.1 A positive dominant residue

Put

$$V = (1-2x)(1-x-y)R, \qquad W = (1-3x)V. \tag{11.1}$$

After these three factors are removed, every remaining active transition uses at least one unit of c. Exact grading of the reduced numerator gives

$$\max(\deg_x + \deg_y - \deg_c) = 3. \tag{11.2}$$

Thus $W_{p,q,k,\ell,m} = 0$ when $p+q > k+3$.

Set

$$\mathcal{B}(c,d,e) = W(1/3, y, c, d, e). \tag{11.3}$$

Sixteen exact irreducible cancellations make $\mathcal{B}$ independent of y. Define

$$\mathcal{C} = \frac{W-\mathcal{B}}{1-3x}. \tag{11.4}$$

Then

$$\mathcal{C}_{p,q,k,\ell,m} = 0 \quad (p+q > k+2), \qquad V = \frac{\mathcal{B}}{1-3x} + \mathcal{C}. \tag{11.5}$$

Theorem 11.1 (Positive $x = 1/3$ residue)**.** *The series $\mathcal{B}(c,d,e)$ is coefficientwise nonnegative. Consequently, if $p+q > k+2$, then*

$$V_{p,q,k,\ell,m} = \begin{cases} 3^p\mathcal{B}_{k,\ell,m}, & q = 0, \\ 0, & q > 0. \end{cases} \tag{11.6}$$

Structured proof.

⟨1⟩1. Exact specialization gives

$$\mathcal{B} = \frac{2cd}{3^8}\frac{3-c}{1-\frac{4}{3}c}\frac{3-cd}{1-\frac{4}{3}cd}\frac{1-d}{1-d-\frac{1}{3}cd}J, \tag{11.7}$$

where all displayed factors are multiplied and

$$J = \frac{H}{(1-\frac{4}{3}cde)(1-de-\frac{1}{3}cde)(1-e-\frac{1}{3}cde)} \tag{11.8}$$

with

$$H = 45 - 36e - 36de - 27cde + 27de^2 + 12cde^2 + 12cd^2e^2 + 4c^2d^2e^2. \tag{11.9}$$

⟨1⟩2. The three outer ratios are nonnegative. For example,

$$\frac{3-c}{1-\frac{4}{3}c} = 3 + \frac{3c}{1-\frac{4}{3}c}, \qquad \frac{1-d}{1-d-\frac{1}{3}cd} = 1 + \frac{\frac{1}{3}cd}{1-d-\frac{1}{3}cd}.$$

The cd ratio is identical to the c ratio after replacing c by cd.

⟨1⟩3. Decompose J exactly:

$$\begin{aligned} J = {} & \frac{9(3-3d+cd)}{(1-d)(1-cd)(3-(3+cd)e)} \\ & + \frac{9(3-4c)(3-4cd)}{(1-c)(1-cd)(3-4cde)} \\ & + \frac{9(3-3d-cd)}{(1-c)(1-d)(3-(3d+cd)e)}. \end{aligned} \tag{11.10}$$

Individual summands are signed; they must be combined coefficientwise.

⟨1⟩4. Put

$$S_{n,r} = \sum_{i=0}^{\min(n,r)} \binom{n}{i} 3^{-i}, \qquad S_{n,-1} = 0, \qquad \alpha_n = (4/3)^n. \tag{11.11}$$

After extracting $e^n c^r d^s$ from (11.10), the three pieces have the exact case description recorded by the verifier: the first is supported on $s \geq r$, the second on $r \geq s \geq n$, and the third on $s \geq n$. Outside $0 \leq r \leq s \leq n$ they cancel using $S_{n,r} = \alpha_n$ for $r \geq n$. Inside that triangle every surviving term is nonnegative.

⟨1⟩5. Hence $J \geq 0$, so every factor in (11.7) is coefficientwise nonnegative and $\mathcal{B} \geq 0$.

⟨1⟩6. Equation (11.5) and the support of $\mathcal{C}$ give (11.6). □

11.2 Why the active theorem does not close the proof

The passive head cone is unimodular. Put

$$\alpha = \ell - m, \qquad \beta = \ell - k, \qquad \gamma = k + m - \ell. \tag{11.12}$$

Then

$$(k, \ell, m) = \alpha(1,1,0) + \beta(0,1,1) + \gamma(1,1,1). \tag{11.13}$$

The three rays are cd, de, and cde. The positive residue has positive terminal coefficients on all three, but the signed correction $\mathcal{C}$ has unbounded negative terminal states. At $x = y = 0$,

$$\mathcal{C}(0,0,c,d,e) = \frac{cd}{1-2cde} - \mathcal{B}(c,d,e). \tag{11.14}$$

Therefore

$$[c^n d^n e^0]\mathcal{C}(0,0) = -[c^n d^n e^0]\mathcal{B} < 0 \quad (n \geq 2), \tag{11.15}$$

and

$$[cd^{n+1}e^n]\mathcal{C}(0,0) = -\frac{2}{81} \quad (n \geq 1). \tag{11.16}$$

These are exact infinite families. They prove that the active residue plus the one-variable passive recurrences does not yield a finite polyhedral induction.

11.3 One complete recession ray repairs

For every $n \geq 1$,

$$[cd^{n+1}e^n]R = \Phi(x,y), \tag{11.17}$$

where

$$\Phi = \frac{x^3(1-x)}{(1-2x)(1-3x)(1-x-y)} + \frac{xy(1-x)}{(1-2x)(1-x-y)} + \frac{y(1-x)}{1-2x}. \tag{11.18}$$

This is coefficientwise nonnegative. The exact domination identity is

$$\mathcal{C}_{de} = (1-2x)(1-x-y)\Phi - \frac{2/81}{1-3x}. \tag{11.19}$$

After division by $(1-2x)(1-x-y)$, the negative term cancels the matching part of the positive residue, leaving Φ. The analogous joint domination along the cd ray remains open and is the simplest visible model of the global missing circuit.

12 One-sided eventual positivity and finite falsifier searches

12.1 A separate one-sided extraction

There is a rational series R_{raw} satisfying

$$[x^p y^q c^k d^\ell e^m]R_{\mathrm{raw}} = A^{(6)}_{p,p+q,k,\ell,m}. \tag{12.1}$$

As a rational function of x, it has 18 distinct linear pole factors. The dominant $x = 1/3$ coefficient is

$$\begin{aligned} C_3 = {} & \frac{1}{3(1-\frac{3}{2}y)} \frac{1-\frac{1}{3}c}{1-\frac{4}{3}c} \frac{1-d}{1-d-\frac{1}{3}cd} \frac{1-\frac{1}{3}cd}{1-\frac{4}{3}cd} \\ & \times \frac{1-e}{1-e-\frac{1}{3}cde} \frac{1-de}{1-de-\frac{1}{3}cde} \frac{1-\frac{1}{3}cde}{1-\frac{4}{3}cde}, \end{aligned} \tag{12.2}$$

where all seven factors are multiplied. Each factor is of the form $(1-u)/(1-\lambda u)$ with $\lambda \geq 1$, or $(1-u)/(1-u-v)$; hence $C_3 \geq 0$.

For each fixed supported passive tuple (q, k, ℓ, m), the $3^p C_3$ term eventually dominates the 2^p and polynomial-growth residues. This gives an effective, slice-wise eventual-positivity theorem. It does not give a uniform cutoff $p \geq 4$, because the threshold depends on the passive tuple.

12.2 Finite evidence, stated as finite evidence

- An exact pair scan through $0 \leq p, q \leq 16$, $k \leq 12$, $\ell \leq 10$, and positive fifth exponent $m \leq 8$ found no negative coefficient of the canonical $\mathcal{E}_6$.
- An exact one-sided scan through $4 \leq p \leq 100$, $q \leq 5$, $k \leq 10$, $\ell \leq 8$, and $m \leq 6$ checked 403,326 coefficients and found no negative.
- Complete full-orbit tables are positive for $\ell = 0, \ldots, 8$, and maximum-charge layers are positive through $p = 6$.

These computations are valuable falsifier searches and regression certificates. None proves an all-degree statement.

13 A universal A_5-kernel lift into A_6

Define

$$\begin{gathered} U_{41} = z_1 + z_4 - z_6, \qquad U_{42} = z_2 + z_4 - z_6, \qquad U_{51} = z_1 + z_5 - z_6, \\ \mathcal{R}_6 = (2z_1 - z_6)(z_1 + z_2 - z_6)(2z_2 - z_6)(z_1 + z_3 - z_6)(z_2 + z_3 - z_6)(2z_3 - z_6), \end{gathered} \tag{13.1}$$

where the six factors in $\mathcal{R}_6$ are multiplied.

Theorem 13.1 (Universal lifted kernel)**.** *Let*

$$L = \alpha(z_1 + z_2) + \beta z_3 + \gamma(z_4 + z_5) + \delta z_6. \tag{13.2}$$

Then

$$\mathcal{K}_L = GBC\, U_{41} U_{42} U_{51} L \tag{13.3}$$

is universally A_6 residue-null: replacing Q_6 by $Q_6 - \mathcal{K}_L$ changes no A_6 Thom polynomial at any level. For $L = -z_6$,

$$[z_6^4](Q_6 - \mathcal{K}_{-z_6}) = GDJ. \tag{13.4}$$

Structured proof.

⟨1⟩1. Cancel the three endpoint-six weights. The nine new D_6/D_5 factors consist of U_{41}, U_{42}, U_{51} and the six factors in $\mathcal{R}_6$. Hence

$$\frac{\mathcal{K}_L}{D_6} = \frac{GBC\, L}{D_5 \mathcal{R}_6}. \tag{13.5}$$

The factor $\mathcal{R}_6$ is independent of z_4, z_5 and invariant under s_{12}.

$\langle 1\rangle 2.$ Split $G = F + M$. For the MBC summand,

$$\frac{MBC}{D_5} = \frac{1}{FE\,\mathcal{P}(z_4)\mathcal{P}(z_5)}, \tag{13.6}$$

where

$$\mathcal{P}(t) = (2z_1 - t)(z_1 + z_2 - t)(2z_2 - t)(z_1 + z_3 - t).$$

Apart from the Vandermonde, the complete integrand is invariant under s_{45}. The Vandermonde is antisymmetric.

$\langle 1\rangle 3.$ Justify the s_{45} contour swap. Take nested radii $R_{i+1} > 10R_i$. Every nonzero pole in z_4, z_5 lies on the z_1, z_2, z_3 scale and is inside both outer contours. No pole is crossed. The residue equals its negative and vanishes.

$\langle 1\rangle 4.$ Treat the FBC summand. Let

$$H_0 = \frac{FBC}{D_5} = \frac{1}{E\mathcal{P}(z_4)\mathcal{P}(z_5)M(z_5)}.$$

Its s_{45}-symmetric part vanishes as above. Put $\mathcal{T}(t) = \mathcal{P}(t)(z_2 + z_3 - t)$. Direct subtraction gives

$$H_0 - s_{45}H_0 = \frac{z_5 - z_4}{E\mathcal{T}(z_4)\mathcal{T}(z_5)}. \tag{13.7}$$

$\langle 1\rangle 5.$ Complete the source multiset. The sources of $\mathcal{T}$ are

$$\{2z_1, z_1 + z_2, 2z_2, z_1 + z_3, z_2 + z_3\},$$

which is s_{12}-invariant. The factors $E, L, \mathcal{R}_6$ are also s_{12}-invariant, while the Vandermonde changes sign.

$\langle 1\rangle 6.$ Justify the s_{12} contour swap. Poles from E and $\mathcal{T}$ lie on outer scales. New roots $z_6/2$, $z_6 - z_2$, $z_6 - z_3$, and their partners lie outside both inner contours because $R_6/2 > R_2$ and $R_6 - R_3 > R_2$. No pole is crossed, so the antisymmetric part vanishes.

$\langle 1\rangle 7.$ Both pieces of $FBC + MBC = GBC$ vanish against every symmetric finite Chern insertion. Passing to arbitrary truncation degree proves all-level residue-nullity.

$\langle 1\rangle 8.$ For $L = -z_6$, the top z_6-term of $U_{41}U_{42}U_{51}(-z_6)$ is z_6^4, so $[z_6^4]\mathcal{K}_{-z_6} = GBC$. Since $[z_6^4]Q_6 = GDJ + GBC$, equation (13.4) follows. □

13.1 Why the positive A_5 boundary gauge does not prove A_6

Let $Q_6' = Q_6 - \mathcal{K}_{-z_6}$. Its $e = 0$ boundary is the coefficientwise-positive GDJ representative. Nevertheless two independent exact chamber engines give

$$A'_{0,2,1,0,1} = 0, \qquad A'_{2,2,1,0,1} = -2. \tag{13.8}$$

Thus its first pair at $(p, q, k, \ell, m) = (0, 2, 1, 0, 1)$ equals -2. The lift transports the positive boundary but makes the positive-e first-pair route false.

14 Gauge dependence and geometric no-go theorems

14.1 Marked pair packets are not intrinsic unmarked numbers

Take the boundary-preserving kernel

$$\mathcal{K} = GBC\,U_{41}U_{42}U_{51}(z_1 + z_2). \tag{14.1}$$

It has $[z_6^4]\mathcal{K} = 0$, its normalized correction is divisible by e, and it changes no complete Thom coefficient. At

$$\beta = (2, 3, 2, 1, 1), \qquad s_1\beta = (1, 3, 2, 1, 1), \tag{14.2}$$

the correction contributes 0 and -2. Canonical Q_6 contributes 7 and 4. Therefore the same marked first-pair coefficient in the gauge $Q_6 + t\mathcal{K}$ is

$$11 - 2t. \tag{14.3}$$

At $t = 6$ it is -1, while the complete 30-word orbit for $\{-1, -1, -1, 0, 1, 2\}$ receives total correction zero.

Corollary 14.1 (Intrinsic-geometry obstruction)**.** *An individual coefficient of the canonical $\mathcal{E}_6$ series is not determined by the unmarked Thom class. In particular it cannot, without extra canonical-gauge or flag data, be an intrinsic characteristic number of the unmarked curvilinear augmentation bundle or a central Jucys–Murphy count.*

Proof. An intrinsic quantity determined only by the Thom class is invariant under universal residue-kernel corrections. Equation (14.3) is not invariant. □

14.2 A pointwise two-state cone cannot work

For an ordered coefficient u and its reversed orientation u^τ, put $S = u + u^\tau$ and $A = u - u^\tau$. An asymmetric denominator transition with ordered weights λ, μ acts by

$$M(\lambda, \mu) = \frac{1}{2}\begin{pmatrix} \lambda + \mu & \lambda - \mu \\ \lambda - \mu & \lambda + \mu \end{pmatrix}. \tag{14.4}$$

The factor pair $1 - e - cdex$, $1 - e - cdey$ supplies

$$M_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad M_- = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{14.5}$$

The exact A_5 boundary contains both $(S, A) = (0, 2)$ and $(0, -2)$. Any position-independent convex cone containing these vectors and invariant under M_+ contains both the antisymmetric line and the diagonal line, hence is all of $\mathbb{R}^2$. It cannot certify the half-space $S \geq 0$. This rules out that precise pointwise two-state architecture, not spatially coupled or history-sensitive cones.

14.3 Other exact obstructions

- Raw Laurent positivity is false; negative ordered coefficients can be repaired only after orbit aggregation.
- The finite 15-factor active core is signed, even at the origin.
- Fixed active strips $p = 0, 1, 2, 3$ have fully reduced signed numerators with respectively 1,144, 10,308, 40,093, and 109,202 terms.
- On an effective square-zero boundary $\mathrm{Gr}(2, 3) \simeq \mathbb{P}^2$, the augmentation quotient has $m_{(2,0)} = -h^2$. Effectivity of a boundary stratum does not imply ordinary Chern-monomial positivity.
- The first nontrivial nested-incidence descendant $S_2^{5,3}$ has no representation by the natural two-dimensional symmetric tautological state: the required three exact linear equations are inconsistent.

Exploratory numerical linear programmes also reject several small positive-atom ansatzes. They are not used as exact no-go theorems because no rational dual certificates were reconstructed.

15 The exact remaining proposition

Conjecture 15.1 (Canonical A_6/A_5 pair-extension positivity)**.** *For the canonical saturated Borel-orbit numerator Q_6,*

$$\mathcal{E}_6(x, y, c, d, e) \in \mathbb{Z}_{\geq 0}[[x, y, c, d, e]]. \tag{15.1}$$

Conjecture 15.1 is sufficient for full A_6 positivity by Theorem 9.2. It is stronger than the final theorem and is gauge-dependent: a universal kernel correction can make one of its marked coefficients negative without changing any Thom coefficient. Therefore failure of Conjecture 15.1, if found, would refute this proof route but would not by itself refute Rimányi's A_6 conjecture.

15.1 The smallest visible domination problem

The exact decomposition

$$(1 - 2x)(1 - x - y)R = \frac{\mathcal{B}}{1 - 3x} + \mathcal{C} \tag{15.2}$$

has a positive first summand and a finite-width active correction. The correction nevertheless contains infinite negative terminal families on the *cd* and *de* rays. The *de* ray repairs by the circuit (11.19); the *cd* analogue is open. A proof of the *cd* domination which is stable under the two remaining ray directions is the most concrete next lemma.

15.2 What would be a proof

Any of the following would be decisive:

1. a coefficientwise-positive decomposition of the canonical $\mathcal{E}_6$ rational function into verified positive kernels;

2. a well-founded recurrence with a rational polyhedral *spatially coupled* invariant cone that contains all exact source vectors and whose output functional is the symmetric pair coefficient;

3. a sign-reversing involution or weighted path model whose unmatched objects enumerate the $\mathcal{E}_6$ coefficients;

4. a canonical-gauge flagged-geometric model producing the marked first-pair coefficients as nonnegative degrees; or

5. a direct full-S_6 argument proving $C_6(M) \geq 0$ without passing through $\mathcal{E}_6$.

Each proposed certificate must be checked symbolically in all degrees. Verification on a large box is not a substitute.

15.3 What would be a counterexample

- A negative coefficient of canonical $\mathcal{E}_6$ is an exact counterexample to Conjecture 15.1 and the current first-swap route, but not necessarily to A_6 Chern positivity.
- A negative first-S_3 packet refutes that coarser sufficient route, but can still be repaired by later folds.
- A negative complete distinct ballot-orbit sum $C_6(M)$ is an exact counterexample to the A_6 case of Rimányi's conjecture, conditional on the certified Q_6 and residue formula.

16 What the new ideas reveal

16.1 The object is globally positive, not locally positive

The basic phenomenon begins at A_5. Ordered chamber coefficients, individual coordinate-prime components, and partial boundary strata can be negative. Positivity appears only after a global balancing operation: complete permutation packets, a chamber-safe contour-kernel relation, or a Stanley circuit that keeps numerator and denominator factors together.

At A_6, there are two simultaneous sources of nonlocality:

1. the raw associativity scheme has an extra top-dimensional component, so the saturated Morin multidegree requires a global component subtraction; and

2. the correct Chern coefficient is a full S_6 orbit packet, while intermediate marked subpackets depend on the chosen numerator gauge.

This explains why componentwise effectivity, ordered Laurent positivity, and a local two-state recurrence all fail for principled reasons.

16.2 Coxeter and 0-Hecke viewpoint

The operators Π_i are the idempotent simple generators of the 0-Hecke algebra of S_6. They sort a charge word while summing coefficients, with wall retention encoding stabilizers. The complete Chern coefficient is therefore the terminal state of a Coxeter-theoretic folding process.

This viewpoint separates two questions:

- *local fold positivity*, which is sufficient but can fail at intermediate stages; and
- *terminal orbit positivity*, the actual theorem.

The exact example in the $p = 2$ layer where a state -2 is later combined with 57 to give 55, and ultimately a positive full orbit, is a small model of this distinction.

16.3 Symmetric-function viewpoint

Introduce an alphabet X and complete symmetric functions $h_j(X)$. The constant-term transform

$$\mathcal{H}_{d,L}(X) = \mathrm{CT}_{\mathcal{C}}\left[(z_1 \cdots z_d)^L K_d(z) \prod_{i=1}^{d} \sum_{j\geq 0} h_j(X) z_i^{-j}\right] \tag{16.1}$$

becomes $\mathrm{Tp}_{A_d}^{L-1}$ after the formal substitution $h_j \mapsto c_j$. Rimányi positivity is thus positivity in the monomial basis of the algebraically independent h_j, not ordinary Schur positivity. The orbit packet formula is the coefficient-level version of (16.1).

16.4 Curvilinear Hilbert schemes and Jucys–Murphy elements

The curvilinear component

$$C_{d+1,r} = \mathrm{Hilb}_0^{d+1,\mathrm{curv}}(\mathbb{A}^r)$$

has a rank-d augmentation bundle E. The colleague report identifies Chern coefficients with integrals of monomial symmetric functions $m_\nu(E)$. Repeated hyperplane descent reduces the sector $\lambda_1 \leq L+1$ to the plane, where Jucys–Murphy elements encode minimal transposition factorizations. This ties the positive sector to Hilbert schemes, the center of the symmetric-group algebra, and Hurwitz-like enumeration.

The gauge-dependence theorem explains the limitation: a marked pair packet is not an unmarked characteristic number. A successful geometric model must remember a flag, chamber, or unresolved collision history.

16.5 Rational generating functions and analytic combinatorics

The dominant $x = 1/3$ pole controls slice-wise eventual positivity. This is an analytic-combinatorial fact made exact by partial fractions: the 3^p contribution eventually dominates 2^p and polynomial-growth terms for each fixed passive tuple. Uniform positivity fails to follow because the residue coefficients and threshold vary with the passive degrees. The signed recession rays show why multivariate asymptotics must be uniform over a cone, not merely pointwise.

16.6 Universal residue kernels as gauge transformations

Equation (4.4) makes numerator choice analogous to a gauge: different fixed-chamber arrays represent the same Thom class. The A_5 proof chooses a gauge with positive ordered coefficients. The new A_6 kernel lift transports that boundary gauge but cannot make the marked positive-e pairs positive. This suggests studying:

- the space of chamber-safe rational gauges, rather than the abstract orbit-sum kernel alone;
- normal forms selected by saturation, minimal poles, or a flag geometry; and
- gauge-invariant positivity certificates acting only after complete orbit folding.

17 Further applications and new questions

17.1 Potential applications of the exact ideas

1. **Higher Morin orders.** The source-multiset completion behind Theorem 13.1 is not specific to one coefficient. It suggests a systematic method for lifting lower-rank kernel relations through endpoint factors in D_{d+1}/D_d.
2. **Certified orbit multidegrees.** The component-plus-Gröbner workflow for Q_6 is reusable whenever an associativity or incidence scheme has extra top components. It separates flat multidegree computation from the scheme-theoretic multiplicity audit.
3. **Positive rational circuits.** The identities (8.8), (10.8), and (11.19) suggest a library of coefficientwise-positive rational circuits larger than the cone generated by monomials over positive denominators. Such a library may be useful for other multivariate positivity problems with inclusion–exclusion numerators.
4. **Flagged Hilbert-scheme invariants.** The no-go theorem points toward characteristic numbers on a flagged or nested curvilinear space, where the chamber order and an unresolved first-S_2 block remain visible.
5. **Computer-assisted exact positivity.** The rational-factor and recurrence certificates provide a test case for automated discovery of invariant cones with exact rational dual certificates.

17.2 Concrete open questions

1. Does the canonical series $\mathcal{E}_6$ have a negative coefficient, or is Conjecture 15.1 true?
2. Can the cd-ray correction in (11.15) be paired with the positive residue by an identity analogous to (11.19)?
3. Is there a finite spatial stencil whose exact transition matrices preserve a rational polyhedral cone, even though no pointwise two-state cone can work?
4. Is there a canonical representative of the universal residue class selected by minimal denominator, saturation, or a geometric flag, and is its full 0-Hecke fold manifestly positive?

5. Can the p-layer characteristic numbers $\int_{C_{7,p+1}} m_\nu(E)$ be interpreted by a positive history-sensitive algebra extending Jucys–Murphy elements beyond the plane descent sector?
6. Do the three passive rays cd, de, cde correspond to three geometric boundary operations on nested curvilinear schemes?
7. Does the source-multiset completion criterion classify all chamber-safe contour-swap kernels generated by lower-rank factors?
8. Can the non-reductive-GIT A_6 blow-up tree be published as an exact leaf table and used as a genuinely independent Q_6 provider?
9. Is full A_6 positivity true even if the canonical first-pair series eventually fails? If so, what is the smallest parabolic or full orbit grouping that is uniformly positive?

18 Reproducibility and audit trail

18.1 Exact commands rerun for this dossier

```
sage research/a6/root_q6/reconstruct_q6.sage
sage -python research/a6/q6_geometry/verify_component_decomposition.py

python3 research/a6/a5_kernel_lift/verify_kernel_lift.py
python3 research/a6/geometric_subpacket_no_go/verify_marked_gauge_obstruction.py
python3 research/a6/reduced_pair/active_residue.py

python3 -m pytest -q research/a6 --ignore=research/a6/raw_cutoff
sage -python -m pytest -q research/a6/raw_cutoff/test_one_sided_tail.py
```

The ordinary Python suite passed 34 tests before the final active-residue additions; the reduced-pair agent then reran its enlarged suite with 29 passing tests. The Sage raw-cutoff suite passed 5 tests in 186.61 seconds. The exact Q_6 reconstruction and prime-decomposition outputs agree with the values in Theorems 5.1 and 5.3.

18.2 Primary artifacts

Artifact	Purpose
`research/a6/root_q6/reconstruct_q6.sage`	Independent Gröbner multidegree reconstruction.
`research/a6/q6_geometry/verify_component_decomposition.py`	Exact prime intersection and multiplicity audit.
`research/a6/computation/data/q6_reconstructed.json`	Validated sparse Q_6 input.
`research/a6/full_orbit_attack/full_orbit_operator.py`	Exact 0-Hecke folding and full orbit sums.

Artifact	Purpose
`research/a6/full_orbit_attack/a5_boundary_dependency.md`	Boundary deletion lemma and conditional completion dependency.
`research/a6/root_q6/first_swap_tail_theorem.md`	Complete singleton proof.
`research/a6/reduced_pair/README.md`	Reduced pair, faces, recurrences, active residue, and proof boundary.
`research/a6/reduced_pair/active_residue.py`	Exact $x = 1/3$ factorization and recession-ray audit.
`research/a6/a5_kernel_lift/README.md`	Universal contour-kernel lift and pole audit.
`research/a6/geometric_subpacket_no_go/README.md`	Gauge-dependence and intrinsic-geometry obstruction.
`research/a6/raw_cutoff/README.md`	One-sided dominant residue and slice-wise eventual positivity.

18.3 Dependency boundary

The exact conclusions depend on:

1. the published Bérczi–Szenes iterated-residue theorem and its fixed chamber;
2. the published geometric identification of the desired $d = 6$ Borel orbit with the non-linear component;
3. the certified ideal, multidegree, and normalization computations;
4. the imported colleague proof for full A_5 packets; and
5. standard exact formal-power-series algebra.

The finite scans additionally depend on implementation correctness inside their declared boxes. Cross-checking two engines reduces, but cannot eliminate, shared upstream errors.

19 Final assessment

The A_6 project has not reached a complete proof or a complete-orbit counterexample. It has reached a substantially sharper point:

- the geometric input Q_6 is no longer opaque;
- every Chern coefficient has an exact stabilizer-safe ballot formula;
- the complete A_5 boundary is used correctly;
- every first-swap singleton is proved nonnegative;
- the only missing statement in that completion route is canonical $\mathcal{E}_6 \geq 0$;

- large infinite sectors and recession rays are proved;
- the dominant active residue is positive;
- the remaining signed head is proved to persist on unbounded rays; and
- several attractive but invalid local or intrinsic arguments are ruled out by exact counterexamples.

The mathematical message is that Morin Chern positivity, if true at A_6, is a property of saturated geometry plus global orbit balancing. It is not positivity of the raw equations, a chosen Laurent series, a single degeneration component, an unmarked boundary number, or an individual local fold. The next proof must encode the cancellation history rather than erase it.

A Logical implication diagram

$$\begin{array}{c}\text{exact stabilization } F_6|_{e=0} = F_5 \\ + \text{ full imported } A_5 \text{ packet theorem}\end{array} \Longrightarrow B_0(M) \geq 0,$$

$$\left.\begin{array}{c}\mathcal{E}_6 \geq 0 \\ \text{singleton theorem}\end{array}\right\} \Longrightarrow B_+(M) \geq 0,$$

$$B_0(M) \geq 0 \quad + \quad B_+(M) \geq 0 \Longrightarrow C_6(M) \geq 0 \Longrightarrow \text{full } A_6 \text{ Chern positivity.}$$

B A compact proof-status table

Statement	Status	Authority
Bérczi–Szenes residue formula	Published theorem	Annals of Mathematics (2012).
Schur positivity in the stable complex setting	Published theorem	Pragacz–Weber and related classifying-space results.
Full A_5 ordinary Chern positivity	Imported theorem	Colleague report; critical identities reconstructed.
Exact 418-term Q_6	Certified	Prime decomposition plus Gröbner multidegree.
Ballot packet and 0-Hecke formulas	Exact	Formal residue and finite-orbit proofs.
Full A_6, $\ell \leq 8$	Exact finite	Complete sharp-box computations.
First-swap singleton theorem	Exact infinite	Triangular support plus positive specialization.
c^1, c^2d, c^2d^2 faces	Exact infinite	Rational decompositions and recurrences.
Positive $x = 1/3$ active residue	Exact infinite	Factorization and coefficient case proof.

Statement	Status	Authority
Universal A_5-kernel lift	Exact infinite	Two chamber-safe contour swaps.
Canonical $\mathcal{E}_6 \geq 0$	Open	No proof and no negative coefficient found.
Full all-level A_6 Chern positivity	Open	Conditional on the preceding open proposition in the present route.

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